Process & Decision Documentation

Project/Assignment Decisions

- At one stage, attempts were made to implement a slash effect that caused the game to freeze ot disappear due to rendering conflicts. Instead of continuing with the complex implementation, I decided to simplif ythe visual effects and start with basic visuals such as the trails, and adjusted theboar knockback distance. Instead of increasing knockbcack strength, the knockback duration(knocktimer) was extended to allow enemies to travel farther
- Another key decision was where sounds should be triggeredin the code. Instead of playing sounds from the input system, sounds were placed at meaningful game play event such as under the space attack key, I would load the swing.mp3.

Side Quests and A4 (Individual Work)

- One key decision I made was that I was choosing which files to modify when adding new features like sound effects and gameplay adjustments. I added and loaded sound effects in the assetloader file which made it organizedand easy to access like “hit enemy”, collectLeaf, and recieveDmage
- Instead of modifying the boat.js to adjust knockback behavior and rewriting the combat system, I increased the knockback duration so enemies travel farther when hit.

Role-Based Process Evidence

```
const wantAttack = input?.attackPressed;
if (
  p.knockTimer === 0 &&
  !p.pendingDeath &&
  grounded &&
  !p.attacking &&
  wantAttack
) {
  p.startAttack();
  this.events?.emit("player:attacked", {});

  // Play swing sound
  const s = this.assets?.sounds?.swing;      You, 34 minutes ago • Unc
  if (s) {
    s.currentTime = 0; // restart sound if it's still playing
    s.play().catch(() => {}); // prevent browser errors
  }
}
```

```
// sprites
this.sprite = null;
this.sensor = null;

// simple visual effect
this.trailTimer = 0; // frames left to draw trail      You,
this.trailPositions = []; // store recent positions

// spawn
const ps = this.levelData.playerStart || {
```

```
// timers      You, 3 days ago • files added …
tickTimers() {
  if (this.invulnTimer > 0) this.invulnTimer--;
  if (this.knockTimer > 0) this.knockTimer--;
  // update trail positions
  // ----------------------
  this.trailTimer++;
  if (this.trailTimer % 2 === 0) {
    // every 2 frames
    // store the current position
    this.trailPositions.push({ x: this.sprite.x, y: this.sprite.y });
    // keep last 6 positions max
    if (this.trailPositions.length > 6) this.trailPositions.shift();
  }
}
```

GenAI Documentation

GenAi was used for this project.

Date Used: March 7, 2026

Tool Disclosure: ChatGPT (GPT-5.3)

Purpose of Use: I used ChatGPT primarily for debugging javascript issues, understanding code errors, and geting guidance on where to place new features in the game files.

Summary of Interaction: The tool helped explain how different parts of the game code functioned and suggested ways to implement features such as sound effects, collision/physics logic for trailing and knockbacks. It also provided examples on how to structure sound playback using the asset loader, and where to place sounds to trigger during gameplay events.

Human Decision Point(s): I decided where the suggested code should actually be placed in my project files and adjusted some of the suggestions to match me existing code structure. For instance, I chose to play the leaf collection sound inside the leaf collection function instead of in the collision check so the ysound would only play when the leaf was successfully collected. I also simplified some suggested implementations when earlier attempts caused the game to freeze.

Integrity & Verification Note:

I verified the suggestions by testing the code directly in the game, checking whether the feature worked as expected (for example, confirming that sounds played at the correct events, or if it even made any sounds) if the game broke, I modified or removed the code until it worked correctly.

Scope of GenAI Use:

The existing code, gameplay systems, and assets were already provided and implemented as part of the sidequest. I mainly used GenAI to debug problems, understand existing systems, and add small features like sound triggers, and gameplay physics adjustments.

Limitations or Misfires: AI could not fully understand the entire game code structure as it was split into multiple working files and sometimes they thought the code was more condensed into one a few js files which made the code much more complex and redundant. Thus, the suggestions causes the game to freeze or screen disppearing. I removed the changes and restrarted from a stable version of the code.

Summary of Process (Human + Tool)

Describe what you did, focusing on process rather than outcome. This may include:

The parts that I implemented mostly myself were loading in the new audio files, increasing the values for knockback and knockback timer, attempting the loop for creating the affect. I tested these small code changes to see if the game still ran, and further asked AI what I was missing and where I went wrong when I did not get the right outcomes. Ai helped debug my trailing effect, and guided where to place codes such as the sound effect for leaf collection and enemy hits.

Decision Points & Trade-offs

Describe one or two key decisions you made:

- I simplified the slash visual effect after it caused the gae to freeze. I ende dup removing the effect due to it not working, and focuses on improving gameplay feedback through sound effects and knockback adjustments
- One sound that was not included was a general “hit” sound effect when pressing space, so I loaded a sound in from pixabay and loaded it in myself. I think it was finethat the swing sound played at the sametime as the “hitenemy” sound as that’s how physics should work.

Verification & Judgement

I I evaluated changes mainly through playtesting the game after each iteration and change. If a feature worked, I kept it. I also checked that sounds triggered at the correct moements and adjusted it if it didn’t. I compared results with assignment requirements to see if I’ve learned and applied what I should have for this sidequest.

Limitations, Dead Ends, or Open Questions

- The last hit for the “hitenemy” sound effect had no sound, I think it triggered as ‘defeat” and not an enemy hit.

Appendix

Prompt (paraphrased):
How do I make a sound play when the player hits an enemy?

GenAI Response (excerpt):
“Load the sound inside your asset loader and trigger it inside the enemy damage logic so it only plays when the enemy actually takes damage.”

Suggested Code:

```
const s = this.assets?.sounds?.hitEnemy;
if (s) {
 s.currentTime = 0;
 s.play();
}
```

How I Used It:
I added the sound to the asset loader and triggered it inside the enemy hit logic. After testing, the sound played correctly when enemies were knocked back.

Prompt (paraphrased):
I placed the sound files in the project folder but they aren't playing. Do I need to load them somewhere?

GenAI Response (excerpt):
"Add the audio files to your asset loader so they are initialized with the rest of the game assets."

Suggested Code:

```
this.sounds = {
 hitEnemy: new Audio("assets/sfx/hitEnemy.wav"),
 collectLeaf: new Audio("assets/sfx/leafCollect.wav"),
 receiveDamage: new Audio("assets/sfx/receiveDamage.wav")
};
```

How I Used It:
I added the sound definitions to the asset loader so they could be accessed from other files using `this.assets.sounds`.

Prompt (paraphrased):
Where should I trigger the sound when a leaf is collected?

GenAI Response (excerpt):
"Place the sound inside the function that handles the leaf being rescued so the sound only plays when the action is successful."

Suggested Code:

```
const s = this.assets?.sounds?.leafCollect;
if (s) {
 s.currentTime = 0;
 s.play();
}
```

How I Used It:
I placed this code inside the leaf rescue function so it plays after the leaf is removed and the score increases.

Prompt (paraphrased):
How can I add a swing sound when the player presses the attack key without affecting the enemy hit sound?

GenAI Response (excerpt):
"Trigger the swing sound when the attack input is detected while keeping the hit sound tied to the enemy damage system."

Suggested Code:

```
const s = this.assets?.sounds?.swing;
if (s) {
 s.currentTime = 0;
 s.play();
}
```

How I Used It:
I added the swing sound to the asset loader and triggered it when the player presses the attack key.

Prompt (paraphrased):
How can I add looping background music to the game?

GenAI Response (excerpt):
"You can load the music like other sounds and set it to loop so it plays continuously during gameplay."

Suggested Code:

```
const bgm = assets.sounds?.music;
if (bgm) {
 bgm.loop = true;
 bgm.volume = 0.4;
 bgm.play();
}
```

How I Used It:
I added the music file to the asset loader and triggered it during the game initialization so it plays during gameplay.

Prompt (paraphrased):
How can I make enemies get knocked back farther when they are hit?

GenAI Response (excerpt):
"You can increase the knockback distance by adjusting the knockback timer or velocity in the enemy entity logic."

Suggested Code Example:

```
e.knockTimer = this.KNOCK_FRAMES * 2;
```

How I Used It:
I modified the knockback values so enemies travel farther when hit, improving the combat feedback.

Prompt (paraphrased):
The jump sound is not playing when I press the jump key even though other sounds work.

GenAI Response (excerpt):
"Make sure the sound trigger is placed inside the part of the code where the jump actually occurs rather than just checking the input."

Suggested Code Example:

```
const s = this.assets?.sounds?.jump;
if (s) {
 s.currentTime = 0;
 s.play();
}
```

How I Used It:
I attempted to add the sound inside the jump logic but later moved on to implementing other sound effects when this feature did not work immediately.

Prompt (paraphrased):
After adding a slash visual effect the game froze and stopped rendering. What might be causing this?

GenAI Response (excerpt):
"The issue could be related to how the effect is being drawn inside the render loop. Try simplifying the effect or removing it temporarily to isolate the problem."

How I Used It:
I removed the slash effect and focused on improving gameplay feedback using sound effects instead.

Prompt (paraphrased):
Why do I need an `if (s)` check before playing the sound?

GenAI Response (excerpt):
"This check ensures the sound exists before calling `.play()` so the game does not crash if the asset failed to load."

Suggested Code:

```
if (s) {
 s.currentTime = 0;
 s.play();
}
```

How I Used It:
I kept this check in the code to prevent errors if a sound file fails to load.

Prompt (paraphrased):
I asked how to add a simple visual trail effect behind the player to make movement look more dynamic.

GenAI Response (excerpt):
"You can store previous player positions and draw them with reduced opacity to create a fading trail effect behind the character."

Suggested Code Example:

```
this.trail.push({ x: player.x, y: player.y });

if (this.trail.length > 10) {
 this.trail.shift();
}
```

And in the draw loop:

```
for (let t of this.trail) {
 ctx.globalAlpha = 0.3;
 ctx.drawImage(playerSprite, t.x, t.y);
}
ctx.globalAlpha = 1;
```

How I Used It:
I added a small trail array to store recent player positions and drew them with lower opacity to create a fading effect. This made player movement feel smoother and added visual feedback when moving quickly.